# Dividing Complex Numbers with Conjugates

*Multiplying a complex number by its conjugate, dividing by a pure imaginary number, dividing by a binomial denominator, and using complex division to find impedance and current in an AC circuit*

**The one move.** A quotient of complex numbers is not in standard form until the denominator is real. Multiply the numerator and the denominator by the *conjugate* of the denominator, which changes the sign of its imaginary part only.

- The conjugate of $a + bi$ is $a - bi$, and $(a + bi)(a - bi) = a^2 + b^2$ — always a real number, never negative.
- Multiplying the top and the bottom by the same number does not change the value of the quotient; it only changes how it is written.
- Finish by dividing *both* parts of the numerator by that real denominator and writing the answer as $a + bi$.
- In circuit work, voltage $V$, current $I$ and impedance $Z$ are complex and obey $V = I \cdot Z$, so a division recovers whichever one is missing. Voltage is in volts, current in amps, impedance in ohms.

**1.** Write the conjugate of $7 + 2i$, then multiply $7 + 2i$ by that conjugate. Give the product.

Answer: ________________

**2.** Write in standard form: $\dfrac{6 + 4i}{2i}$

The denominator is pure imaginary, so its conjugate is $-2i$.

Answer: ________________

**3.** Write in standard form: $\dfrac{5+5i}{2+i}$

Answer: ________________

**4.** Write in standard form: $\dfrac{11+10i}{4-i}$

Answer: ________________

**5.** An AC circuit carries a current of $3+2i$ amps while the voltage across it is $2+23i$ volts. Impedance is defined by $Z = V/I$. Find the impedance of the circuit in ohms, written in standard form.

**Answer:** ________________ ohms

**6.** Write in standard form: $\dfrac{11 - 10i}{3 + 2i}$

Priya divided the real parts and the imaginary parts separately. Say in one sentence why that method does not work here, then do it correctly.

Answer: ________________

**7.** Write in standard form: $\dfrac{9 - 6i}{3i}$

Answer: ________________

**8.** Write in standard form: $\dfrac{25}{3 + 4i}$

The numerator is already real. State what the denominator times its conjugate equals before you simplify.

Answer: ________________

**9.** A speaker with impedance $2 + i$ ohms is driven by a voltage of $1 + 8i$ volts. Ohm's law for AC circuits says $V = I \cdot Z$. Find the current $I$ through the speaker in amps, written in standard form.

**Answer:** ________________ amps

**10.** **Synthesis challenge.** Each part is one division in disguise.

**(a)** Solve for $x$: $(3 + 2i)\,x = 4 + 7i$ _________

**(b)** Find the zero of $f(x) = (1 - i)\,x - (6 + 2i)$ _________

In one sentence, say which conjugate you multiplied by in part (b) and why.